

IT 123 - Lab Report 2

Ubuntu User, Group, and Permission Management

Scope: Ubuntu Server only, as directed by the student. No Windows Server VM was available. The Windows portion of the course guide remains outstanding. Student name and section were not supplied.

1. Objective

Create local users and groups, assign directory access, and verify that permitted operations succeed while unauthorized operations fail.

2. Environment and setup

Existing VirtualBox guest: Ubuntu 26.04.1 LTS. The administrator account kadmin was already available. The setup script checked that target users, groups, and directories did not exist, then created student2, faculty2, and student4 with home directories. Passwords were set through hidden prompts and are not recorded in this report.

Account / group	Directory / ownership	Mode
student2 / labusers	/labdata root:labusers	770, then 750
faculty2 / facultygrp	/facultydata root:facultygrp	770
student4 / studentgrp	/studentdata root:studentgrp	750

The student2 account comment is "Test account for Week 3 Lab". Each directory contains a root-owned sample file. The final /labdata mode is 750: group members can read and traverse, but cannot write. Its initial mode 770 allowed the required before-and-after write test.

3. Procedure

The setup script created accounts and groups, assigned supplementary groups, and applied root ownership and modes. The verification script ran commands under each new user identity, recorded exit codes, and changed /labdata from 770 to 750 between the initial and final checks. Separate password-backed `su` sessions verified that each account could authenticate.

The complete commands are in scripts/setup_ubuntu.sh and scripts/verify_ubuntu.sh. The original test output is in verification-output.txt.

4. Verification results

Result: 12 access checks passed, 0 failed, script exit code 0. Each of the three password-backed account logins also succeeded.

Test	Expected	Observed
student2 reads /labdata at 770 and 750	Allow	PASS / exit 0
student2 creates a file at 770	Allow	PASS / exit 0
student2 creates a file at 750	Deny	PASS / permission denied
faculty2 reads and writes /facultydata	Allow	PASS / exit 0
student4 reads /studentdata	Allow	PASS / exit 0
student4 creates or edits in /studentdata	Deny	PASS / permission denied
Cross-group directory access	Deny	PASS / permission denied
student2, faculty2, student4 logins	Allow	PASS / correct user IDs

5. Interpretation

The before-and-after /labdata test demonstrates that removing group write permission blocks new files while preserving read access. The separate faculty and student directories allow their designated groups the requested access. Cross-group attempts fail. Password-backed `su` logins validate authentication; `runuser` tests in the script validate file authorization.

6. Evidence and limitations

The following screenshots were captured directly from the Ubuntu VM after setup and testing. The raw test log records the initial and final /labdata modes, allowed operations, and permission-denied errors. The login tests were observed in the live terminal session and summarized here; separate login screenshots were not captured. Windows Server work is outside this Ubuntu-only report.

Source: six student-provided photographs of the IT 123 Week 3 laboratory guide and rubric. The guide includes Windows Server tasks as well as Ubuntu tasks.

Appendix A - User and group evidence

```
uid=1001(student2) gid=1001(student2) groups=1001(student2),100(users),1004(labusers)
uid=1002(faculty2) gid=1002(faculty2) groups=1002(faculty2),100(users),1005(facultygrp)
uid=1003(student4) gid=1003(student4) groups=1003(student4),100(users),1006(studentgrp)
student2:x:1001:1001:Test account for Week 3 Lab:/home/student2:/bin/bash
faculty2:x:1002:1002::/home/faculty2:/bin/bash
student4:x:1003:1003::/home/student4:/bin/bash
kadmin@it123:~$ _
```

Figure 1. User IDs, supplementary groups, home directories, and the student2 comment. Screenshot cropped to its terminal output; the full capture is in screenshots/.

Appendix B - Directory permission evidence

```
drwxr-x--- 750 root:labusers /labdata
drwxrwx--- 770 root:facultygrp /facultydata
drwxr-x--- 750 root:studentgrp /studentdata
ls: cannot open directory '/facultydata': Permission denied
ls: cannot open directory '/labdata': Permission denied
ls: cannot open directory '/studentdata': Permission denied
kadmin@it123:~$ _
```

Figure 2. Final ownership and modes. kadmin receives Permission denied when listing the restricted directories. Screenshot cropped to its terminal output; the full capture is in screenshots/.

Appendix C - Access-test evidence

```
PASS | student2 | read lab sample at 750 | exit=0
touch: cannot touch '/labdata/denied-test.txt': Permission denied
PASS | student2 | create lab file at 750 | exit=1
Week 3 permission verification sample
PASS | faculty2 | read faculty sample | exit=0
PASS | faculty2 | create faculty file | exit=0
Week 3 permission verification sample
PASS | student4 | read student sample | exit=0
touch: cannot touch '/studentdata/denied-test.txt': Permission denied
PASS | student4 | create student file | exit=1
sh: 1: cannot create /studentdata/sample.txt: Permission denied
PASS | student4 | modify existing student sample | exit=2
cat: /facultydata/sample.txt: Permission denied
PASS | student4 | read faculty sample | exit=1
cat: /studentdata/sample.txt: Permission denied
PASS | faculty2 | read student sample | exit=1
cat: /facultydata/sample.txt: Permission denied
PASS | student2 | read faculty sample | exit=1
cat: /studentdata/sample.txt: Permission denied
PASS | student2 | read student sample | exit=1
Failed checks: 0
Now verify passwords through interactive logins; runuser does not test them.
kadmin@it123:~$
```

Figure 3. Successful and denied operations from the Ubuntu test run, ending with zero failed checks. Screenshot cropped to the test output; the full capture is in screenshots/.